Microcontrollers: sense, decide, act

IoT · Level 2 · Lesson 4



Today we will

- Name the parts of a microcontroller board: power, pins, inputs and outputs.
- Explain how a program's loop reads a sensor, decides and acts (a night light).
- Choose a way to start: a simulator, a real board, or unplugged.

Read, decide, act

- Inputs bring news in
- Outputs act
- Setup once, then loop
- The rule is what you design

Be the microcontroller

1. Warm-up (5 min): Name devices that switch themselves on and off.
2. Teach (8 min): Inputs, outputs and the loop; the night-light rule.
3. Act it out (10 min): A pupil runs the program card with a torch.
4. Flowchart (12 min): Groups draw the loop and change the threshold.
5. Wrap-up (5 min): Show it in a simulator, if you can.

Write the program card before class: READ the window light; DECIDE: is it darker than our level?; ACT: torch on or off; WAIT 5 seconds; REPEAT. Try MakeCode on your own phone first: the micro:bit's light level runs from 0 to 255, so use a threshold such as 50.

Let’s discuss

1. Which devices around you run a loop like this?
2. Why is it useful that the threshold is a number we can change?
3. What could go wrong if the light sensor sits in the shade?

Exit ticket

- Draw the night light's loop in four boxes, with the rule in the decide box.